

eXpOS:

An Experimental Operating System from India



An operating system is also a program that includes over millions of lines of source code. A cursory glance over the Internet tells us that a typical Linux operating system like Ubuntu or Debian has over 12 million lines of source code. But this enormous volume of source code poses a huge challenge to a person teaching the concepts of operating systems. Developed in India, eXpOS is an educational operating system that addresses this challenge.

How does a teacher approach the task of dividing an operating system into meaningful sub-units and providing useful information about them to the students? Of course, there are some operating systems that are very small, with just a few lines of source code. For example, KolibriOS is an operating system with a GUI (graphical user interface) that requires only about 2MB of disk space and less than 10MB of RAM to run. So, is it wise to use such a tiny, lightweight operating system to teach the concepts? Yes, it is. Some academicians believe such an approach is the right one. An operating system called MINIX (a UNIX-like OS) is an excellent example

for this approach, wherein a real, yet minimal operating system is used to teach the concepts. Older versions of MINIX were developed by Dr Andrew S. Tanenbaum of Vrije Universiteit Amsterdam in the Netherlands, solely for educational purposes. But from MINIX 3 onwards, the objective has changed to include the creation of a highly reliable and secure micro-kernel operating system.

But there is also a second school of thought on teaching various concepts in computer science, in general, and operating systems in particular. People who support this second school of thought believe in using abstract entities to teach concepts in computer science. An excellent example is a project called 'From Nand to Tetris' developed by Dr Noam Nisan (Hebrew University of Jerusalem) and Dr Shimon Schocken (IDC Herzliya, Israel). This project involves the development of a simulated computer called 'Hack' starting from simulated NAND gates, and is licensed under